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Khadka, Anamol, "Credit Card Fraud Detection via Model Retraining and Fine-Tuning" (2025). *Computer Science and Engineering Student Research*. 1.

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Credit Card Fraud Detection via Model Retraining and Fine-Tuning

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Abstract—Credit card fraud detection is a critical task in financial systems, especially given the rarity and evolving nature of the fraudulent behavior. The highly imbalanced class levels of the fraudulent and non-fraudulent transactions make it a challenging classification problem to solve. This study investigates the effectiveness of machine learning models: Logistic Regression, XGBoost, and Multi-Layer Perceptron (Neural Network), evaluated under temporal retraining and fine-tuning scenarios using a publicly available, highly imbalanced dataset of European credit card transactions. The dataset includes 284,807 transactions, of which only 492 (0.172%) are labeled as fraudulent, making it a well-known example of an imbalanced classification problem. The models are evaluated using metrics including precision-recall AUC, ROC-AUC, and class-specific precision and recall—emphasizing capturing minority fraud cases. The research indicated that temporal retraining preserves or improves recall over time, while fine-tuning enables neural networks to adapt incrementally without the requirement of full training. Among the three models, XGBoost demonstrates the best balance of precision and recall across all setups. These findings highlight the value of continual learning strategies in real-world fraud detection pipelines.

Index Terms—Credit Card Fraud, Fraud Detection, Unbalanced Data, Machine Learning, Concept Drift, Temporal Validation, Fine-tuning, Imbalanced Learning.

I. INTRODUCTION

Credit card fraud continues to pose significant risks to financial institutions, merchants, and consumers. With the rapid increase in online transactions and digital payment platforms, fraudulent activities have become more sophisticated, making detection increasingly challenging. The inherently imbalanced nature of fraud datasets—where fraudulent transactions represent only a tiny fraction of total activity—further complicates the task for traditional and machine learning-based classification models.

This study focuses on evaluating and enhancing the performance of machine learning models for credit card fraud detection using a publicly available dataset collected in Europe. The dataset contains 284,807 transactions, among which only 492 are labeled as fraudulent, resulting in a fraud rate of approximately 0.172%. The data includes principal components (V1–V28) obtained through PCA transformation, along with two additional features: Amount, indicating the transaction value, and Time, representing the seconds elapsed since the first transaction. The binary Class label identifies each transaction as legitimate (0) or fraudulent (1).

Given the class imbalance and the evolving nature of fraudulent patterns, this work investigates not only traditional

model performance on randomly partitioned data but also temporal retraining, where models are evaluated on later transactions after being trained on earlier ones. In addition, the study explores fine-tuning strategies, simulating real-world conditions where models must adapt incrementally without being retrained from scratch.

By comparing logistic regression, gradient-boosted decision trees (XGBoost), and a neural network-based multilayer perceptron (MLP), this research provides insights into how different models respond to evolving fraud patterns and which strategies best maintain high recall and precision over time. The results contribute toward building robust, adaptable fraud detection systems suitable for real-time financial environments.

II. RELATED WORK

Prior research has extensively addressed fraud detection using machine learning [1]. Imbalanced learning techniques [2] and concept drift adaptation [3] have emerged to address evolving fraud patterns. Recent work also investigates active learning in streaming settings [4], highlighting challenges in labeling and model updates.

Credit card fraud detection poses several inherent challenges, including extreme class imbalance, evolving fraud patterns (dataset shift), lack of sequential data, and the need for model interpretability. Lucas and Jurgovsky [1] provide a foundational survey of these challenges and categorize various machine learning techniques proposed to address them. The authors begin by outlining the structure of typical fraud detection datasets, emphasizing that performance evaluation must go beyond simple accuracy and instead use metrics like precision, recall, and PR-AUC. They also explore how class imbalance affects both data sampling strategies and model behavior, summarizing approaches such as cost-sensitive learning and resampling. A central focus of the survey is on dataset shift, or concept drift, where fraudulent behavior evolves over time and models must be updated or fine-tuned to remain effective. The paper also reviews feature engineering and sequence modeling methods, including LSTM networks and Markov models, to improve temporal awareness in fraud detection systems. While not proposing new algorithms or experiments, this work serves as a comprehensive starting point, highlighting relevant techniques and papers for each major subproblem in the fraud detection pipeline.

Class imbalance is a central challenge in credit card fraud detection and many real-world machine learning tasks where

the minority class carries disproportionately high importance. Chen et al. [2] present a comprehensive and modern survey of imbalanced learning techniques across both classical and deep learning contexts. The authors categorize imbalance-handling strategies into five key areas: data-level methods (e.g., SMOTE, ADASYN), algorithm-level approaches (e.g., cost-sensitive learning), hybrid techniques, ensemble models, and emerging paradigms such as long-tail learning and imbalance in streaming data. Unlike most surveys, this paper extends the discussion beyond the binary classification task to include multiclass, regression, and semi-supervised machine learning facets. The paper also highlights the limitations of using accuracy as an evaluation metric in skewed datasets and advocates for more informative metrics such as PR-AUC, ROC-AUC, and F1-score. Furthermore, it identifies future research directions in interpretability, data generation, and online adaptation. As such, the paper provides both a theoretical and practical foundation for understanding and addressing class imbalance in dynamic fraud detection systems.

Dal Pozzolo et al. [3] present a framework tailored to real-world credit card fraud detection systems (FDSs) that must operate under two core challenges: severe class imbalance and the presence of concept drift over time. The study emphasizes that fraud detection must account for the delayed availability of ground truth labels, where only a small subset of risky transactions are promptly verified by investigators while the rest are labeled days later by customer reports. The authors propose that these two sources of supervision, feedback from investigators and delayed labels, should be handled separately instead of being pooled together, as traditionally done. They introduce and evaluate two learning strategies: sliding window and ensemble-based methods. Their approach involves training two separate classifiers, one on recent feedbacks and another on delayed labels, and combining them to enhance precision. Experimental evaluations on real-world datasets reveal that the proposed hybrid models (AWt and AEt), which treat feedback and delayed data independently, significantly outperform conventional models in terms of alert precision. This work provides critical insights into designing fraud detection pipelines that are robust to non-stationarity and verification latency, which are key challenges in operational FDSs.

Carcillo et al. [4] explore the design of credit card fraud detection systems that operate in streaming environments, where data arrives continuously, and labels are both limited and delayed. To address the challenges of class imbalance, concept drift, and constrained labeling budgets, the authors investigate a set of active learning (AL) strategies that prioritize which transactions should be reviewed by human experts. Their study categorizes these strategies into highest-risk querying (HRQ), exploratory active learning (EAL), and stochastic semi-supervised learning (SSSL). Through experiments on a real-world dataset of over 12 million transactions, they demonstrate that combining risk-based querying with semi-supervised labeling of low-risk transactions significantly improves fraud detection performance. The authors also introduce a visualization framework to track how the learning

process adapts over time, helping to balance exploitation (focusing on known fraud) and exploration (discovering new fraud patterns). This work reinforces the importance of adaptive learning systems in real-time fraud detection and provides evidence that targeted sampling and limited supervision can rival fully labeled approaches. Although streaming or active learning is not implemented directly in this study, similar principles are adopted through temporal retraining and fine-tuning to improve adaptability to evolving fraud patterns.

III. DATASET AND PREPROCESSING

This study utilizes the publicly available Kaggle Credit Card Fraud Detection dataset [5], which comprises 284,807 anonymized transaction records collected over two days by European cardholders. Among these transactions, only 492 are labeled as fraudulent, reflecting a severe class imbalance with a fraud prevalence of approximately 0.172%. Each transaction is characterized by 30 features, including 28 anonymized principal components (V1 to V28) derived through Principal Component Analysis (PCA). The PCA transformation was applied by the data provider to preserve confidentiality while reducing dimensionality and mitigating feature correlation. In addition to these components, the dataset includes two non-anonymized features: Amount, indicating the transaction value in euros, and Time, representing the elapsed seconds since the first transaction.

Initial preprocessing confirmed that the dataset contains no missing or null values, negating the need for imputation. The Amount and Time features, which were not subject to PCA, were standardized using z-score normalization via the `StandardScaler` to ensure consistent input scales for downstream models. Importantly, the natural class imbalance was preserved during training to simulate real-world deployment scenarios, and mitigation strategies were incorporated at the modeling stage using class-weighted loss functions or algorithm-specific parameters.

To support robust model evaluation, two data splitting strategies were employed. First, a stratified random split was applied to preserve class distribution between training and testing sets, enabling baseline comparisons under controlled conditions. Second, a temporal split was introduced by sorting the dataset chronologically using the Time feature. In this configuration, the first 80% of the data was used for training and the last 20% for testing, simulating a more realistic fraud detection scenario where models trained on historical data must generalize to future, potentially evolving patterns. This preprocessing framework laid the groundwork for evaluating baseline models, temporal retraining, and fine-tuning strategies in the subsequent experiments.

IV. METHODOLOGY

To investigate the adaptability and robustness of fraud detection models under evolving data conditions, this study employs three distinct modeling approaches: a linear classifier (Logistic Regression), a tree-based ensemble model (XGBoost), and a neural network (Multilayer Perceptron, MLP). These models

are first evaluated using standard random splits to establish baseline performance, followed by evaluation under temporally split data to simulate real-world deployment conditions. Finally, a fine-tuning protocol is applied to assess the models' ability to adapt incrementally to newer fraud patterns without full retraining.

A. Baseline Models

- **Logistic regression:** This model serves as a simple, interpretable linear classifier. To address the extreme class imbalance, the `class_weight="balanced"` option is used, allowing the model to assign higher penalty to misclassified fraud cases. Features are standardized using z-score normalization through a `StandardScaler` pipeline.
- **XGBoost:** The gradient-boosted tree model is implemented using the `XGBClassifier` from the XGBoost library. This model is particularly well-suited for handling imbalanced data via the `scale_pos_weight` parameter, which accounts for the ratio of negative to positive samples. It is trained with 300 trees, a learning rate of 0.05, and a maximum depth of 5, with subsampling to enhance generalization.
- **Multilayer Perceptron (MLP):** A neural network model is constructed using PyTorch, consisting of two hidden layers with 64 and 32 neurons, respectively, and ReLU activation. Dropout regularization is applied to prevent overfitting. The model uses `BCEWithLogitsLoss` with a `pos_weight` parameter to handle class imbalance and is trained using the Adam optimizer over 10 epochs.

B. Temporal Splitting Strategy

To simulate a more realistic fraud detection environment, the dataset is chronologically sorted based on the Time feature. The first 80% of the transactions are used for training, while the remaining 20% constitute the temporal test set. This approach mimics deployment scenarios where models must generalize to future, potentially unseen fraud patterns. All three models are trained using this split to assess their robustness to concept drift.

C. Fine-Tuning Protocol

A fine-tuning experiment is conducted to evaluate each model's ability to adapt incrementally without full retraining. The dataset is partitioned into three sequential segments:

- Base training set (first 60%)
- Fine-tuning set (next 20%)
- Final test set (last 20%)

Each model is initially trained on the base set. The XGBoost model is then fine-tuned using additional boosting rounds on the intermediate set, while the MLP continues training for five additional epochs using the new data. Logistic Regression is excluded from fine-tuning due to its static training mechanism. This setup allows for an assessment of how models retain prior knowledge while integrating new patterns from recent transactions.

V. EVALUATION METRICS

Given the extreme class imbalance present in the dataset—where fraudulent transactions account for only 0.172% of the total—a careful choice of evaluation metrics is essential. Conventional accuracy is not a suitable measure in this context, as it may misleadingly report high performance due to the dominance of legitimate transactions. Instead, this study adopts metrics that specifically focus on the minority class and are more informative for fraud detection systems. The following metrics are used to evaluate model performance:

- **Precision (Fraud Class):** Precision quantifies the proportion of transactions predicted as fraudulent that are truly fraudulent. High precision indicates fewer false positives, which is critical for reducing unnecessary alerts in real-world systems. It is defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

- **Recall (Fraud Class):** Recall measures the proportion of actual fraudulent transactions that are correctly identified by the model. High recall is vital to ensure that most fraud cases are caught. It is calculated as:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

- **F1-Score:** The harmonic mean of precision and recall, the F1-score provides a balanced metric when both false positives and false negatives are costly. It is especially informative when comparing models with different trade-offs.
- **ROC-AUC (Receiver Operating Characteristic - Area Under the Curve):** This metric assesses the model's ability to distinguish between classes across all classification thresholds by plotting the true positive rate against the false positive rate. Although informative, it can sometimes be overly optimistic in highly imbalanced settings.
- **PR-AUC (Precision-Recall Area Under the Curve):** Given the class imbalance, PR-AUC is the most representative measure for this task. It summarizes the tradeoff between precision and recall across different thresholds and provides a reliable assessment of performance on the minority class.

All metrics are computed using `scikit-learn` implementations, and evaluation is performed on the respective test sets defined by random and temporal splits. Emphasis is placed on PR-AUC and recall, as these best reflect the ability of models to detect rare but critical fraudulent activity.

VI. EXPERIMENTS AND RESULTS

This section presents the outcomes of model evaluations under three different settings: baseline performance using random splits, temporal retraining to simulate concept drift, and fine-tuning to adapt models incrementally. Each model was assessed using PR-AUC, ROC-AUC, precision, and recall—metrics appropriate for highly imbalanced classification tasks such as fraud detection.

A. Baseline Evaluation

Three baseline classifiers were developed using random 80/20 train-test splits: Logistic Regression (LogReg), Gradient Boosted Trees (XGBoost), and a Multilayer Perceptron (MLP). Class imbalance was addressed using model-specific techniques: `class_weighting` in LogReg, `scale_pos_weight` in XGBoost, and `pos_weight` with `BCEWithLogitsLoss` in MLP. All models were trained on scaled features using z-score normalization.

XGBoost achieved the highest PR-AUC (0.8322) and precision (0.8864), suggesting a strong ability to correctly identify fraudulent cases while minimizing false positives. The MLP model showed the highest recall (0.8980), although with lower precision. Logistic Regression exhibited balanced recall with modest precision, forming a solid benchmark for interpretability-focused scenarios (see Fig. 1 on the next page).

B. Temporal Retraining Simulation

To assess how models perform when trained on older data and tested on future transactions, the dataset was temporally split: the first 80% of time-ordered transactions served as the training set, and the remaining 20% as the test set. This mimics production settings where models must generalize to evolving fraud strategies.

All models demonstrated improved or preserved performance compared to their baseline versions. Notably, the MLP improved both PR-AUC and precision under temporal splitting, indicating stronger adaptability to shifting fraud patterns. Logistic Regression maintained high recall and improved PR-AUC, while XGBoost continued to perform consistently, although with a slight decrease in recall (see Fig. 2 on the next page).

C. Fine-Tuning Evaluation

To simulate an incremental learning scenario, models were trained on the first 60% of the data, fine-tuned on the next 20%, and evaluated on the final 20%. This approach reflects real-world applications where models are updated periodically rather than retrained from scratch.

Fine-tuned XGBoost achieved strong performance (PR-AUC: 0.7873, Precision: 0.6264), effectively balancing adaptability with robustness. The fine-tuned MLP also demonstrated excellent recall (0.8667) and improved PR-AUC (0.7635), although precision remained modest. These results affirm that fine-tuning can enhance detection of newer fraud patterns without overfitting to older data (see Fig. 3 on the next page).

VII. DISCUSSION

The experimental results reveal important insights about the behavior and adaptability of different machine learning models in the context of credit card fraud detection. The baseline evaluation demonstrated that XGBoost outperformed both Logistic Regression and MLP in terms of overall precision and AUC metrics, highlighting its robustness against class imbalance. Logistic Regression, while less precise, achieved high recall, indicating a conservative approach that favors

identifying more fraud cases at the cost of false positives. MLP showed competitive recall but weaker precision due to the inherent challenges of training neural networks on imbalanced datasets without extensive tuning.

Temporal retraining simulation highlighted how model performance evolves over time. All three models preserved or slightly improved their performance when evaluated on temporally forward data. This suggests that the models generalize reasonably well even under distributional shifts. MLP, in particular, showed the greatest improvement in PR-AUC and precision, indicating that its neural architecture is well-suited to learn from evolving patterns when trained in a temporally-aware setting.

Fine-tuning provided further performance enhancement. XGBoost maintained strong recall and precision through incremental training, demonstrating the effectiveness of continual learning without full retraining. The MLP model also benefited significantly, improving its PR-AUC and recall, reinforcing its potential in real-time or streaming fraud detection applications.

However, limitations exist. First, the dataset used—though realistic—is synthetic and only spans two days. This limits the assessment of true long-term concept drift. Second, neural networks, while promising, require more careful hyperparameter tuning and may be sensitive to initialization and training dynamics. Lastly, the results depend on the choice of evaluation metrics; in highly imbalanced settings, precision-recall metrics are more informative than accuracy, but may still overlook business-critical trade-offs.

In real-world deployment, periodic retraining or fine-tuning appears critical, especially in fraud environments that evolve rapidly due to changing user behavior and attacker strategies. The study affirms the relevance of temporal validation, robust modeling strategies like XGBoost, and lightweight fine-tuning techniques as practical strategies for building adaptable fraud detection systems.

VIII. CONCLUSION

This project explored the effectiveness of various machine learning approaches for credit card fraud detection under realistic, imbalanced, and evolving conditions. Three baseline models—Logistic Regression, XGBoost, and a PyTorch-based MLP—were evaluated using both standard and temporally split datasets. The results established XGBoost as the most robust model, excelling in precision and AUC metrics, while MLP demonstrated strong adaptability under temporal shifts and fine-tuning.

Temporal retraining simulations confirmed that models trained on earlier data could maintain or improve performance on newer transactions, validating the importance of evaluating models under shifting data distributions. Fine-tuning simulations further demonstrated that models can incrementally adapt to new data without full retraining, an essential capability for real-time fraud detection systems.

The project contributes a reproducible pipeline that combines practical ML techniques with temporal and continual learning strategies. These findings can guide practitioners in

Baseline Model Comparison

Model	PR-AUC	ROC-AUC	Precision (Fraud)	Recall (Fraud)
Logistic Regression	0.7067	0.9713	0.0653	0.8776
XGBoost	0.8322	0.9838	0.8864	0.7959
MLP (Neural Net)	0.6911	0.9748	0.0724	0.898

Fig. 1. Baseline model performance across key evaluation metrics

Model Comparison: Baseline vs Temporal

Model	PR-AUC (Baseline)	ROC-AUC (Baseline)	Precision (Baseline)	Recall (Baseline)	PR-AUC (Temporal)	ROC-AUC (Temporal)	Precision (Temporal)	Recall (Temporal)
Logistic Regression	0.7067	0.9713	0.0653	0.8776	0.762	0.9863	0.0707	0.8933
XGBoost	0.8322	0.9838	0.8864	0.7959	0.7933	0.9841	0.8261	0.76
MLP (Neural Net)	0.6911	0.9748	0.0724	0.898	0.8049	0.9907	0.1475	0.8533

Fig. 2. Temporal retraining evaluation of models using time-based data split

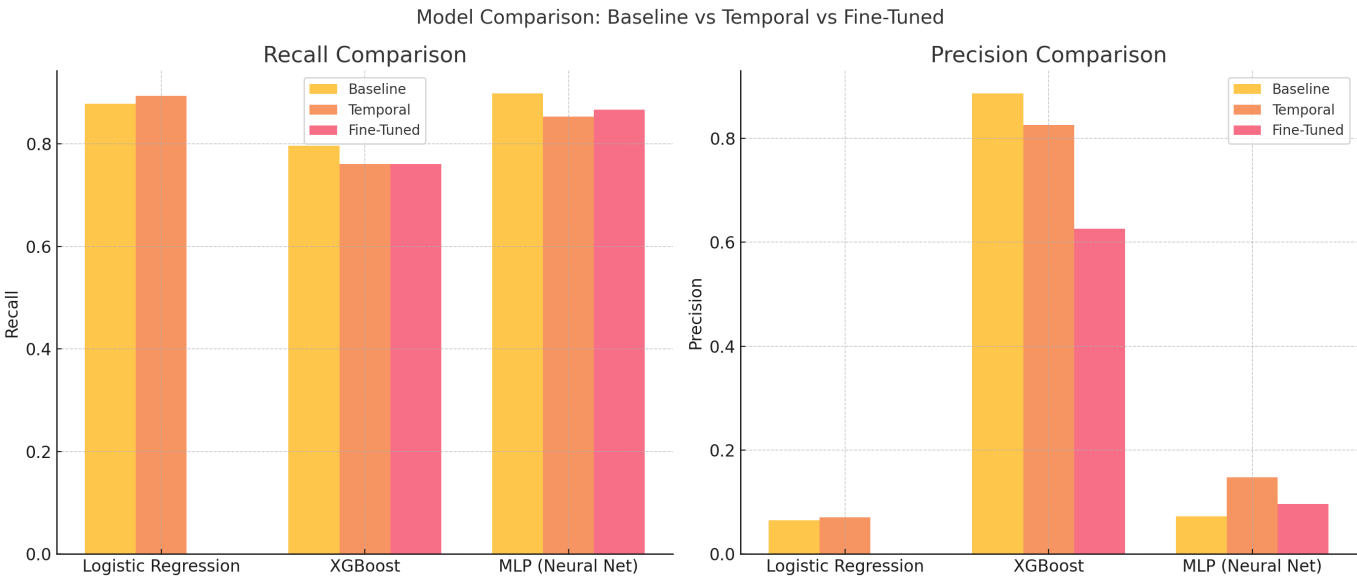


Fig. 3. Baseline, Temporal and Fine-Tuned models comparison for recall and precision

designing scalable, adaptive fraud detection systems capable of handling the complexities of real-world financial data.

Project Repository: The source code and notebook for this work are available at https://github.com/anamolkhadka/creditCard_Fraud

IX. FUTURE WORK

Future research could extend this work by:

- Incorporating more sophisticated neural architectures (e.g., LSTM or Transformers) for sequential pattern detection.
- Testing of the models on longer-duration datasets to evaluate long-term concept drift.
- Exploring real-time data streams and active learning strategies to reduce labeling costs.

REFERENCES

- [1] Y. Lucas and J. Jurgovsky, "Credit Card Fraud Detection Using Machine Learning: A Survey," *arXiv preprint arXiv:2010.06479*, 2020.
- [2] W. Chen, K. Yang, Z. Yu, et al., "A survey on imbalanced learning: latest research, applications and future directions," *Artificial Intelligence Review*, vol. 57, p. 137, 2024.
- [3] A. Dal Pozzolo, G. Boracchi, O. Caelen, C. Alippi, and G. Bontempi, "Credit Card Fraud Detection and Concept-Drift Adaptation with Delayed Supervised Information," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, 2015, pp. 1–8.
- [4] F. Carcillo, Y.-A. Le Borgne, O. Caelen, and G. Bontempi, "Streaming Active Learning Strategies for Real-Life Credit Card Fraud Detection: Assessment and Visualization," *arXiv preprint arXiv:1804.07481*, 2018.
- [5] ULB Machine Learning Group, "Credit Card Fraud Detection," Kaggle Dataset, 2016.